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NATURAL RESOURCES: Forest resources -use and over exploitation –Water resources -use and over utilization - dams - benefits and problems - water conservation -Energy resources - growing energy needs - renewable and non-renewable energy sources - use of alternate energy sources: role of an individual in conservation of natural resources.

Natural Resources

- Introduction to Natural Resources, Forest Resources-Uses and over exploitation
- Water Resources-Uses and over exploitation, Dams-Benefits and Problems
- Water Conservation, Energy Resources-Growing Energy Needs
- Renewable and Non-Renewable Energy Sources, Use of Alternative Energy Sources
- Role of an Individual in the Conservation of Natural Resources

Introduction

- Natural resources are materials and substances in the environment essential for human survival and economic development. They include air, water, soil, minerals, forests, wildlife, and fossil fuels.
- Proper management and conservation of natural resources are crucial to ensure their availability for future generations and to mitigate negative environmental impacts.
- They provide the raw materials and energy needed for human activities and are a crucial foundation of the global economy.
- However, the increasing demand for natural resources and their unsustainable use has led to environmental degradation and resource depletion.

Importance of Natural Resources

(Recall the values of biodiversity from previous

The importance of natural resources can be summarized as follows:

1.Economic Importance: Natural resources are essential for producing goods and services that drive economic growth and development. For example, agriculture depends on fertile soil and water resources, while manufacturing relies on mineral and energy sources.

2.Ecological Importance: Natural resources are fundamental to the functioning of ecosystems and support the planet's biodiversity. They provide habitat for wildlife, regulate climate, and support nutrient and water cycles.

3.Social Importance: Natural resources provide recreation and inspiration, providing cultural and spiritual values to many communities. For example, forests, mountains, and oceans are important for tourism and recreation.

4.Health Importance: Natural resources like clean air and water are essential for human health and well-being. Poor environmental quality can lead to health problems, including respiratory diseases, waterborne illnesses, and exposure to toxins.

Types of Natural Resources?

1.Renewable: resources that are available in infinite quantity and can be used repeatedly are called renewable resources. Example: Solar, wind, hydropower, biomass

2.Non-Renewable: resources that are limited in abundance due to their non-renewable nature and whose availability may run out in the future are called non-renewable resources. Examples include fossil fuels, minerals, etc.

Difference between Renewable and Non-Renewable Resources

Renewable resource	**Non-renewable resource**
It can be renewed as it is available in infinite quantity	Once completely consumed, it cannot be renewed due to limited stock
Sustainable in nature	Exhaustible in nature
Low cost and environment-friendly	High cost and less environment-friendly
Replenish quickly	Replenish slowly or do not replenish naturally at all

Most Important Natural Resources

1.Water: Freshwater is essential for human survival, agriculture, and industry. It is a finite resource and is becoming increasingly scarce in some regions of the world.

2.Air: Clean air is essential for human health and the health of ecosystems, and it is also important for climate regulation and precipitation formation.

3.Fossil Fuels: Oil, natural gas, and coal are the most widely used energy sources for transportation and electricity generation. They are non-renewable and have negative environmental impacts but remain essential to many industries and economies.

4.Minerals: Metals, such as iron, copper, and gold, are essential for industry and technology. Non-metal minerals, such as salt and limestone, are also important for construction and agriculture.

5.Forests: Forests provide numerous ecological, economic, and social benefits, including carbon sequestration, biodiversity conservation, and timber production.

Uses of Natural Resources

1.Food production: Natural resources such as fertile soil, water, and sunlight are essential for food production. Agricultural resources, such as crops and livestock, provide food for humans and animals.

2.Energy production: Natural resources such as coal, oil, and natural gas produce energy for electricity, transportation, and heating. Renewable energy resources such as wind, solar, and hydro are also increasingly used to meet energy needs.

3.Raw materials: Natural resources are used for many products, such as construction materials, metals, plastics, and textiles. For example, timber from forests is used to make furniture, and metals such as iron and copper are used to make machinery and electronics.

4.Recreation and tourism: Natural resources such as forests, beaches, and mountains are popular destinations for recreation and tourism. They provide opportunities for activities such as hiking, camping, and wildlife viewing.

5.Cultural and spiritual values: Natural resources have cultural and spiritual values for many communities. They are often used in cultural practices, such as ceremonies and rituals, and are considered sacred by some groups.

Forest Resources

- Forest resources play an important role in the economy of any country. It is highly complex, changing environment made up of a living and non living things. Living things include trees, shrubs, wildlife etc. and non-living things include water, nutrients, rocks, sunlight and air.

- The word forest is derived from a Latin word " Foris" means Outside
- Forest are one of the most important natural resources of the earth.
- Approximately 1/3rd of the earth's total area is covered by forests

Forests are essential to the global ecosystem and provide a variety of environmental, economic, and social benefits. Timber is the most valuable forest resource, providing lumber, paper, and other wood products.

Indian Scenario

In India forest cover Overall, 21.02% of the country's geographical area is now under green cover **(as per 2009* data)** The total forest cover in India is **6,90,899 km^2**

Forest cover in India is defined as all lands, more than one hectare in area with a tree canopy density of more than 10%.

Very Dense Forest	All lands with tree cover of canopy density of 70% and above
Moderately Dense Forest	All lands with tree cover of canopy density between 40% and 70%
Open Forest	All lands with tree cover of canopy density between 10% and 40%.
Scrub	Degraded forest lands with canopy density less than 10 %.
Non-forest	Any area not included in the above classes.

Functions of Forests

The functions of forest may broadly be classified into following categories

- ***Protective Function*** (Forest Provide protection against Soil erosion, Droughts, floods, noise, radiations)
- ***Productive Function*** (Forest Provide various products like, gum resins, medicines, honey, pulp, bamboo, timber, and fruits)
- ***Regulative Function*** (The Forest regulates the level of Oxygen and carbon dioxide in atmosphere. The forests also help in regulating temperature conditions)
- ***Accessory Function*** (Forest provides aesthetics, habitat to various flora and fauna besides that it also has an recreational value)

Uses of Forest Resources

(i) **Protective Uses:** They serve as wind breakers and also protect the soil from erosion

(ii) **Provision of food:** They provide meat, vegetables, fruits for other living species

(iii) **Medicinal Resources:** Many trees and herbal plants serve as a source of valuable medicines

(iv) **Source of Timber:** Forest is a great source of timber materials which are utilized in the construction of houses, bridges, furniture and vehicles.

(v) **Provision of Raw Materials:** Local and international industries need several raw materials from forests such as, woods, ropes, gums, fibers etc., for manufacturing variety of products

(vi) **Eco tourism:** Forests are useful as tourist attraction and wild park which helps for the income generation of any country

(VII) Employment opportunity: forest provide employment opportunity for many people such as, workers in tourist centres, forest guards, sculptors, hunters etc.

(VIII) Sources of Revenue: By issuing licences for various timber contractors, researchers, hunters, forests can serve as good source of revenue for the governments

(ix) Helpful resource for educational and research activities:

Forest is a great resource for ecological scientists, wild life analysts, botanists, zoologists and students.

(x) Increases the soil fertility and serves as home for wild animals:

Forest trees increases the soil fertility due to the accumulation of humus. It is also a natural home for all kind of animals, micro organisms (fungi, algae, etc.)

Over Exploitation of Forest Resources

Forests are critical natural resources that provide a wide range of ecosystem service, including carbon sequestration, water regulation, soil conservation, and biodiversity conservation. However, forests are also subjected to over-exploited and degradation due to human activities such as, logging, agriculture, and urbanization.

1. **Timber extraction:** Forests are often logged for their valuable timber, which is used for building materials, paper production, and other purposes. Over-logging can lead to deforestation and habitat loss for wildlife.

2. **Agricultural expansion:** Forests are often cleared for agricultural purposes, such as the production of palm oil, soybeans, and other commodities. This can lead to soil degradation, erosion, and loss of biodiversity.

3. Urbanization: Forests are often cleared for urban development, including the construction of roads, buildings, and infrastructure. This can lead to fragmentation of forest habitats and loss of biodiversity.

4. **Fuelwood collection:** Many people in developing countries rely on forests for fuelwood, which can lead to over-exploitation and degradation of forests.

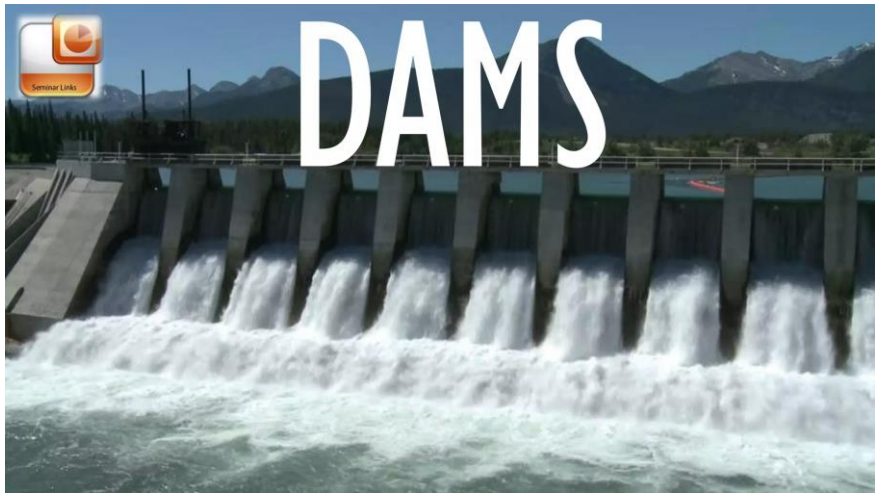
5. Hunting and poaching: illegal hunting and poaching of wildlife for bushmeat and other purposes can lead to over-exploitation and loss of biodiversity.

DAMS

Dam is a solid barrier constructed at a suitable location across a river valley to store water

Storage of water is utilized for following objectives:

- Hydropower
- Irrigation
- Water for domestic consumption
- Drought and flood control
- For navigational facilities
- Other additional utilization is to develop fisheries



Benefits of Dams

The primary purpose of dams is for storage and safe retention of water in large quantity which is subsequently released to achieve various purposes such as irrigation, hydropower, recreation, water supply, flood prevention etc.

Irrigation

The primary and main purpose of constructing a dam is to store water and use it for irrigation. The World Commission of Dam estimates around 30-40% of irrigated land relies on dams. It is estimated that 60% of the food production that comes to the market is dependent on dam water for their irrigation.

Hydropower Generation

Hydropower generated by the dams is the most plentiful and most efficient renewable energy resource, contributing to 24% of all renewable electric energy produced all over the world.

Flood Prevention

Flood control is a significant purpose for many of the existing dams and continues as the main purpose for some of the major dams of the world currently under construction. The dams are used to effectively regulate the level of water flowing in the river by temporarily storing the flood volume and releasing it later.

Water Supply

The major percentage of rainfall precipitates on the ocean and the rainfall on the earth is wasted as runoff, a mere 2% of infiltrated to replenish the groundwater. So properly planned and designed dams can store contribute significantly toward fulfilling our water supply requirements.

Inland Navigation

Dams just not only serve as an irrigational lifeline, but also provide inland navigation throughout the heartland of the nation. The advantages of inland navigation, however, when compared with highway and rail are the large load-carrying capacity of each barge, the ability to handle cargo with large-dimensions and fuel savings.

Recreation

Dams provide prime recreational facilities such as boating, skiing, camping, picnic areas, and boat launch facilities are all supported by dams. The large water stored due to the construction of dams facilitates the growth of flora and fauna in the region due to which many recreational activities such as natural history, bird watching, landscape painting, walking, and hiking are on the rise.

Problems of Dams

Large dams have led to the extinction of many fish and other aquatic species, the disappearance of birds in floodplains, huge losses of forest, wetland and farmland, erosion of coastal deltas, and many other unmitigable impacts.

The problems can be categorized into three categories:

Social Problems:

1.The construction of high-rise dams results in a vast area of land submerged underwater.

2.The people create a displacement of a large number of peasants and tribals without adequate compensation or rehabilitation.

3.The other social problems are generated due to mismanagement by the officials.

4.It has been observed that the people close to the source grow water-intensive crops like sugarcane and rice whereas people farther downstream do not get sufficient water to grow these crops.

5.This also causes social problems resulting due to unequal distribution of water.

Economic Problems:

The construction of a high-rise dam is an expensive project that involves the spending of a huge amount of public money.

Environmental Problems:

1.The construction of high-rise dams causes several problems including deforestation, loss of biodiversity and displacement of poor tribals.

2.The large reservoirs of water formed by the dams destroy plants and animals in the area which get submerged underwater.

Water Conservation

Water conservation is the practice of using water efficiently to reduce unnecessary water usage.

Only 1% of the freshwater is actually accessible to mankind for our use as 97% of water is present in oceans.

Water conservation refers to

- Preservation of water and its resources through careful planning, control, and management
- Activities and strategies undertaken to protect the water
- Efficient use of water by avoiding unnecessary water wastage and usage.

Reasons behind Water Scarcity

- Climate Change and deforestation
- An increase in population has increased the water demands and thus the usage of water by people.
- Increase in pollution in the water bodies through various human activities like industrialization.
- Competition between various industries, cities, and agriculture for acquiring various water resources

Measures for Water Conservation

1. Water Conservation in Domestic Uses

•**Use water judiciously**: Do not overuse water for washing clothes and dishes; close the taps when not in need.

•Check for any **leaks** in your toilets or kitchen pipes and fix them immediately

•Take **showers** with a minimal quantity of water required.

•Use a broom to clean your house instead of washing floors.

•Turn off the tap water while brushing your teeth and shaving.

•Wash your vehicles with a minimum quantity of water required.

• Use mugs to water your plants instead of using flowing water pipes.

2. Methods of Irrigation that Conserve Water

(a) **Drip Irrigation** – Drip irrigation is a type of micro-irrigation system that has the probability of saving water and nutrients by allowing water to drip slowly directly to the roots of plants.

(b) **Sprinkler Irrigation** – Water is distributed through a pipe system, and is sprayed into the air and irrigates in most of the soil types as it has a wide range of discharge capacity.

(c) **Night Irrigation** – During night-time, humidity is high, which results in a reduced evaporation rate. Since there is no sun, solar radiation does not contribute to water evaporation, which minimises water loss.

3. Modern Methods of Water Conservation

Modern water conservation methods are more accessible and more effective, such as water conservation methods at home and on the farm.

(a) **Rainwater Harvesting**

(b) **Protection of water from pollution**

(c) **Wastewater Recycling by Sewage Treatment Plants**

Benefits of Water Conservation

Below are Some of the Important Benefits of Water Conservation

It minimizes the effects of drought and water shortages.

By reducing the amount of water we use, we can better protect against future drought years.

Protects against various conflicts:

Failing to conserve water can eventually lead to a lack of an adequate water supply, which can have drastic consequences. These include rising costs, reduced food supplies, health hazards, and political conflict.

It helps to preserve our environment.

Reducing our water usages reduces the energy required to various processes which, in turn, helps to reduce pollution and conserve fuel resources.

It Serves the recreational purposes.

Much of our freshwater resources are also used for beautifying our surroundings—watering lawns, trees, flowers, and vegetable gardens, filling public fountains at parks.

It builds safe and beautiful communities:

Firefighters, hospitals, gas stations, street cleaners, health clubs, gyms, and restaurants all require large amounts of water to provide services to the community. Reducing our usage of water now means that these services can continue to be provided.

Energy Resources

Energy resources can be defined as materials or elements that can be used to produce energy.

Three primary types of energy include fossil fuels, nuclear energy and renewable energy

Fossil fuel:
Dead organic matter, mostly composed of bacteria, algae and plants, subjected to high heat and pressure over millions of years.

Elemental Sources:
Usually present as major replenishable components of Earth's abiotic spheres.

- Solar
- Wind
- Hydro
- Geothermal

Nuclear: atoms interacting to produce vast quantities of energy

Biomass: plants, algae, bacteria, animals, etc.

Growing Energy Needs

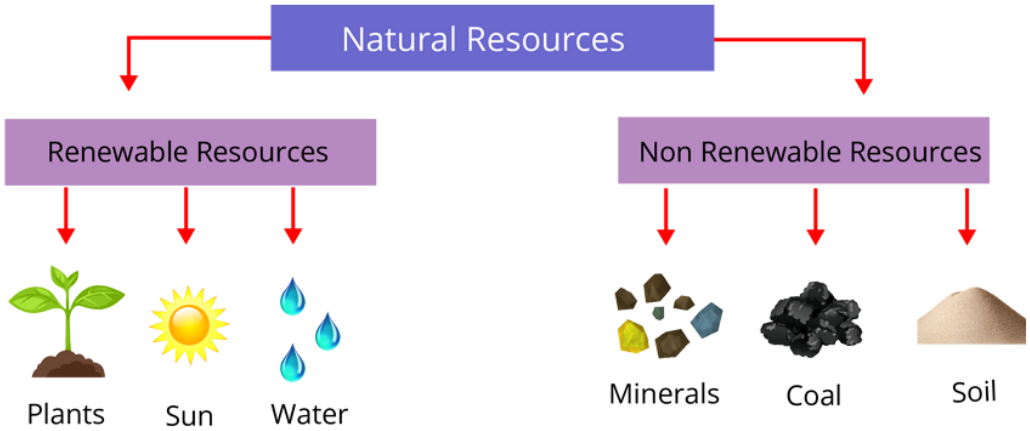
- **Modern agriculture uses chemical fertilizers, which require large amounts of energy during their manufacture**
- Various gadgets in household sector require to use both renewable and non-renewable energy.
- **Industry uses energy to manufacturing processes.**
- Energy demanding roads and railway lines are built to transport products from place to place and to reach raw materials in mines and forests.
- **Almost 2 billion people will require electrical energy and those who have access will continue to increase their individual requirements.**
- Between 1900-2000 world energy consumption has increased by a factor of 14 times while the production has increased only by 3 times.
- **Fossil fuels (eg: coal, oil and gas) were used in 19th and 20th century. Today's energy options include fossil fuels, nuclear energy, solar energy and renewable fuels.**
- World-wide oil demand has raised more than twice for domestic and industrial utilities

Renewable and Non-Renewable Energy Sources

There are two types of natural resources such as, Renewable and Non-Renewable Sources

- **Renewable Resources** which can be generated continuously in nature and are inexhaustible e.g. wood, solar energy, wind energy, tidal energy, hydropower, biomass energy, bio-fuels, geo-thermal energy and hydrogen.

- **Non-renewable Resources** which have formed in nature over a long period of time and cannot be quickly replenished when exhausted e.g. coal, petroleum, natural gas and nuclear fuels like uranium.



Difference between Renewable and Non-renewable Sources of Energy

Renewable	Non-renewable
The resources that can be renewed once they are consumed are called renewable sources of energy.	The resources that cannot be renewed once they are consumed are called non-renewable sources of energy.
These resources do not cause any environmental pollution.	Thèse ressources cause environnemental pollution..
Renewable resources are inexhaustible.	Non- Renewable resources are exhaustible.
Renewable resources are not affected by human activities.	Non- Renewable resources are affected by human activities.
Examples of Renewable resources- Air, water and solar energy.	Examples of Non-renewable resources- natural gas, coal and nuclear energy.

Characteristics of Renewable Energy Sources:

1.Renewable resources have the capacity to restore themselves easily.
2.Ecological developments for a definite time period are required by renewable resources to regenerate themselves.
3.Renewable energy can become non-renewable energy if it is utilised in an extreme amount, that is, higher than the environment's ability to restore it.

Characteristics of Non-Renewable Energy Sources:

1.Non-renewable energy sources are also known as stock resources because they are not obtainable in high quantities.
2.Non-renewable energy generally exists in the form of minerals which are present in various forms in the lithosphere of the earth.
3.Non-renewable resources can be obtained in solids, liquids or gases, that is, all the three states of matter, for instance, coal, petroleum and natural gas.

Renewable Resources : Examples

- **Solar** - The energy obtained from sunlight is solar energy. The sun is the ultimate natural resource for all living beings on the earth. Plants utilise solar energy and make their own food through photosynthesis.
- **Wind** - It is an important renewable resource required for the survival of living organisms. Air is important to carry out photosynthesis and respiration in plants and animal
- **Water**-Hydro energy is generated by water flowing into a river or water held in a dam. Hydro energy is power that is generated from moving water such as rivers.
- **Soil**- Soil is a valuable resource as it is the layer in which plants grow. Living beings require food to live. Plants produce most of the food that is required by living organisms.
- **Biogas**- Biogas is a form of fuel that is a mixture of gases such as methane, carbon dioxide, hydrogen, and other gases produced by the breakdown of animal and plant wastes

Non-renewable Resources : Examples

•**Fossil Fuels-** Fossil fuels are non-renewable energy sources. This means that they will ultimately be finished, which is why energy prices are rising. Fossil fuels consist of coal, natural gas and petroleum.

•**Coal-** Coal is used as a fuel, to generate electricity, and in factories and steam engines.

•**Natural gas-** Natural gas, often known as compressed natural gas, is an excellent alternative to petrol and diesel. It burns quickly and generates a large amount of heat. It's an excellent source of hydrogen.

•**Petroleum-** Mineral oil or crude oil are other names for petroleum. Petrol, diesel, cooking gas, and kerosene are all made from this liquid mineral. It can be found deep within the earth.

•**Nuclear energy-** This energy source involves use of radioactive material that is found in nature. Uranium is primarily used to make nuclear reactor fuel rods. Heat is generated when neutrons(neutral particles present in the atom) hit with the fuel rods. This converts water to steam, which is used to move turbines. As a result, it generates electricity.

Use of Alternative Energy Sources

What are Alternative Sources of Energy?

Alternative sources of energy can be defined as the use of sources of energy other than the traditional fossil fuels (such as oil, coal, and natural gas), which are shorter in supply and which are considered harmful to the environment. It includes all renewable and nuclear energy sources.

The most commonly used alternative sources of energy include the following:

1. Wind Energy
2. Solar Energy
3. Geothermal Energy
4. Bioenergy
5. Hydroelectric Energy
6. Hydrogen Energy
7. Nuclear Energy
8. Tidal Energy

1. Wind Energy

•In this source of energy, the wind is used for producing electricity by making use of the kinetic energy created by air in motion.

2. Solar Energy

•In this source of energy, the radiation from the Sun is used which is capable of producing heat, causing chemical reactions, and generating electricity.

3. Geothermal Energy

•It is a source of energy that is taken from the Earth's core. It comes from the heat which is generated during the original formation of the planet.

4. Bioenergy

•In this source of energy electricity and gas are generated from the organic matter known as biomass. Bioenergy is one of the resources available to help meet the demand for energy, it includes electricity, heat and transportation fuel.

5. Hydroelectric Energy

It is a source of energy that is obtained by falling water from high potential to low potential as potential is defined to flow down from a certain height. It is a form of energy that harnesses the power of water in motion—such as water flowing over a waterfall—to generate electricity.

Importance of Alternative Sources of Energy

1. **Protects the Environment**: Alternative sources of energy help to protect the environment and give it the opportunity to regenerate.

2. **Helps in Providing Sustainable Fuel Systems:** The alternative sources of energy can help in creating a sustainable fuel system and can help in sustaining the ecological balance of a region.

3. **Helps in Reducing the Dependence on Imported Fuels:** Another use of alternate sources of energy is that it helps to reduce the dependence on imported fuels.

4. **Helps in Enhancing Income:** It helps in enhancing the income of the country by creating additional employment opportunities for the population of the country.

5. **Useful in Conserving Fossil Fuels:** When we make use of alternate sources of energy it helps us in conserving fossil fuels.

6. **Useful in Slowing and Reversing Climate Change:** Because the alternate sources of energy have a much lower carbon content it helps to slow down and reverse climate change.

7. **Useful in Economic Growth:** Producing more utility-scale energy systems can help in creating more economic growth.

Role of an Individual in the Conservation of Natural Resources

As global energy demand continues to grow, it is increasingly important for individuals to take an active role in energy conservation. Here are some ways in which individuals can conserve energy:

1. **Use energy-efficient appliances:** Upgrading to energy-efficient appliances, such as LED light bulbs, Energy Star-rated refrigerators and washing machines, can significantly reduce energy usage.
2. **Reduce, reuse, recycle**: Reducing waste, reusing items, and recycling can save energy and resources.
3. **Reduce electricity usage**: Simple steps like turning off lights and electronics when they're not in use, using natural light instead of artificial light, and unplugging chargers when they're not in use can help conserve energy.
4. **Improve insulation**: Improving the insulation in your home can help reduce heating and cooling costs.
5. **Drive less**: Using public transportation, carpooling, biking, or walking instead of driving can save energy and reduce greenhouse gas emissions.
6. **Use renewable energy**: Consider investing in renewable energy sources, such as solar or wind power, to reduce dependence on non-renewable energy sources.
7. **Educate others:** Share your knowledge and encourage others to conserve energy.

Thank You